COMPUTER NETWORKS LAB VIVA GUIDE

Complete question bank with short, exam-ready answers

Topics covered	TCP/UDP • C sockets • Wireshark • HTTP • Packet Tracer • CRC • OSI • Linux
Revision style	Short definitions, function purposes, command examples and tricky viva questions

How to use this guide

Read the bold question, answer aloud in one or two sentences, then try to explain the related program or command. Prioritize socket function sequences, TCP vs UDP, Wireshark filters, HTTP headers, router commands, CRC and bit stuffing.

Prepared for last-minute practical viva revision.

1. TCP and UDP

1. What is a computer network?

Answer: A collection of interconnected devices that communicate and share resources.

2. What is a protocol?

Answer: A set of rules used for communication between devices.

3. What is TCP?

Answer: Transmission Control Protocol; a connection-oriented, reliable transport-layer protocol.

4. What is UDP?

Answer: User Datagram Protocol; a connectionless transport-layer protocol with no delivery guarantee.

5. Differentiate TCP and UDP.

Answer: TCP is connection-oriented, reliable, ordered and has a minimum 20-byte header. UDP is connectionless, has no guaranteed delivery or ordering, and has an 8-byte header.

6. Which is faster, TCP or UDP?

Answer: UDP generally has less communication overhead and can be faster.

7. Give examples of TCP applications.

Answer: HTTP/1.1, HTTP/2, FTP, SMTP and SSH.

8. Give examples of UDP applications.

Answer: DNS (commonly), DHCP, online gaming and VoIP.

9. What is a connection-oriented protocol?

Answer: A protocol that establishes a connection before transferring data.

10. What is a connectionless protocol?

Answer: A protocol that sends data without first establishing a connection.

11. What is the TCP three-way handshake?

Answer: Connection establishment: client sends SYN, server replies SYN-ACK, and client sends ACK.

12. What is the TCP four-way handshake?

Answer: The usual connection termination process using FIN and ACK messages in both directions.

13. What is a TCP segment?

Answer: A unit of data at the TCP transport layer.

14. What is a datagram?

Answer: A unit of data commonly associated with UDP or IP.

15. What is flow control?

Answer: It prevents a sender from sending data faster than the receiver can handle.

16. What is congestion control?

Answer: It prevents excessive traffic from overwhelming the network.

17. What is an ACK?

Answer: An acknowledgement indicating successful receipt of data or control information.

18. What is retransmission?

Answer: Sending data again when it is lost or not acknowledged.

19. What is a socket?

Answer: An endpoint used for communication between processes over a network.

20. What is a port number?

Answer: A number that identifies a particular application or service on a device.

2. C Socket Programming

21. What is socket programming?

Answer: It allows two processes to communicate over a network.

22. Which headers are commonly used for C socket programming on Linux?

Answer: stdio.h, stdlib.h, string.h, unistd.h, sys/socket.h, netinet/in.h and arpa/inet.h.

23. What does socket() do?

Answer: Creates a new socket. Example: socket(AF_INET, SOCK_STREAM, 0).

24. What is AF_INET?

Answer: It specifies the IPv4 address family.

25. What is SOCK_STREAM?

Answer: It specifies a stream socket, commonly used with TCP.

26. What is SOCK_DGRAM?

Answer: It specifies a datagram socket, commonly used with UDP.

27. What is the third argument of socket()?

Answer: It specifies the protocol. Zero selects the default protocol for the chosen socket type.

28. What does bind() do?

Answer: Assigns a local IP address and port number to a socket.

29. What does listen() do?

Answer: Marks a TCP socket as passive so it can accept incoming connection requests.

30. What does accept() do?

Answer: Accepts a client connection and returns a new connected socket descriptor.

31. What does connect() do?

Answer: Allows a client to establish a connection with a server.

32. What does send() do?

Answer: Sends data through a socket.

33. What does recv() do?

Answer: Receives data through a socket.

34. What does read() do?

Answer: Reads data from a file descriptor, including a socket.

35. What does write() do?

Answer: Writes data to a file descriptor, including a socket.

36. What does close() do?

Answer: Closes a file descriptor.

37. What does htons() do?

Answer: Converts a 16-bit value, such as a port number, from host byte order to network byte order.

38. What does htonl() do?

Answer: Converts a 32-bit value from host byte order to network byte order.

39. What does ntohs() do?

Answer: Converts a 16-bit value from network byte order to host byte order.

40. What does ntohl() do?

Answer: Converts a 32-bit value from network byte order to host byte order.

41. What does inet_pton() do?

Answer: Converts an IP address from readable presentation format to binary network format.

42. What does inet_ntop() do?

Answer: Converts a binary IP address into readable text format.

43. What is struct sockaddr_in?

Answer: A structure used to store IPv4 socket address information.

44. What are the important members of sockaddr_in?

Answer: sin_family stores the address family; sin_port stores the port; sin_addr stores the IPv4 address.

45. What is INADDR_ANY?

Answer: It lets a server bind to all available local IPv4 interfaces.

46. What is SO_REUSEADDR?

Answer: A socket option that allows local address reuse in certain situations, often useful when restarting a server.

47. What is a socket descriptor?

Answer: An integer returned by socket() that identifies the socket.

48. What does socket() return on failure?

Answer: It returns -1.

49. What does accept() return on success?

Answer: A new socket descriptor for the connected client.

50. Why is perror() used?

Answer: It prints a descriptive error message based on errno.

51. What is fork()?

Answer: A system call that creates a child process.

52. Why use fork() in a concurrent server?

Answer: To handle clients in separate processes.

53. What is a concurrent server?

Answer: A server that can handle multiple clients without making every client wait for the previous one to finish.

54. What is the TCP server function sequence?

Answer: socket() → bind() → listen() → accept() → send/recv or read/write → close().

55. What is the TCP client function sequence?

Answer: socket() → connect() → send/recv or write/read → close().

56. Difference between listen() and accept()?

Answer: listen() makes the socket passive; accept() accepts a connection request.

57. Why does a server use bind() but a client usually does not?

Answer: A server needs a known local address and port. The OS can assign an available local port to a client automatically.

58. Why use htons() for port numbers?

Answer: Network protocols use network byte order, which is big-endian.

59. Difference between send() and sendto()?

Answer: send() is commonly used on a connected socket; sendto() specifies a destination address and is commonly used with UDP.

60. Difference between recv() and recvfrom()?

Answer: recv() receives data; recvfrom() can also return the sender's address.

61. Why does UDP not use listen() and accept()?

Answer: UDP is connectionless and does not establish TCP-style connections.

62. Can TCP carry multiple messages over one connection?

Answer: Yes, but TCP provides a continuous byte stream and does not preserve message boundaries.

63. Is TCP message-oriented or byte-oriented?

Answer: TCP is byte-oriented.

3. UDP Socket Programming

64. What is the UDP server function sequence?

Answer: socket() → bind() → recvfrom() → sendto() → close().

65. What is the UDP client function sequence?

Answer: socket() → sendto() → recvfrom() → close().

66. Why does UDP use sendto()?

Answer: The destination address is specified because UDP does not establish a connection.

67. Does UDP guarantee packet delivery?

Answer: No.

68. Does UDP perform error checking?

Answer: Yes. UDP has a checksum for error detection, but it does not provide retransmission.

69. What is the size of the UDP header?

Answer: 8 bytes.

70. What is the minimum size of a TCP header?

Answer: 20 bytes.

71. Can UDP be used for file transfer?

Answer: Yes, but reliability mechanisms must be implemented separately if required.

72. Can TCP and UDP use the same port number?

Answer: Yes. TCP and UDP have separate port spaces.

4. Wireshark

73. What is Wireshark?

Answer: A network protocol analyzer used to capture and inspect network packets.

74. What is packet capturing?

Answer: Collecting network packets for analysis.

75. What is a network interface?

Answer: A connection point through which a device communicates, such as Wi-Fi or Ethernet.

76. What is a packet?

Answer: A formatted unit of data transmitted over a network.

77. What is a packet filter?

Answer: A filter used to select packets matching specified conditions.

78. Difference between capture filters and display filters?

Answer: Capture filters select packets during capture and use BPF syntax. Display filters select packets from an existing capture and use Wireshark display-filter syntax.

79. What is the Wireshark display filter for HTTP?

Answer: http

80. What is the Wireshark display filter for TCP?

Answer: tcp

81. What is the Wireshark display filter for UDP?

Answer: udp

82. How do you filter by IP address?

Answer: ip.addr == 192.168.1.10

83. How do you filter by TCP port?

Answer: tcp.port == 80

84. What is a source IP address?

Answer: The IP address of the device that sent the packet.

85. What is a destination IP address?

Answer: The IP address of the device receiving the packet.

86. What is a source port?

Answer: The port used by the sending application.

87. What is a destination port?

Answer: The port used by the receiving application.

88. What does the Protocol column show?

Answer: The protocol Wireshark identifies for a captured packet.

89. What is packet length?

Answer: The size of the captured packet in bytes.

90. What is the Info column?

Answer: A short description of the packet's contents or activity.

91. Difference between a frame and a packet?

Answer: A frame belongs to the data link layer; a packet generally refers to the network layer.

92. What is Follow TCP Stream?

Answer: It reconstructs TCP data exchanged between endpoints into a readable stream.

93. What is a TCP SYN packet?

Answer: A packet used to initiate a TCP connection.

94. What is a TCP ACK packet?

Answer: A packet that acknowledges received data or connection-control information.

95. What is TCP retransmission in Wireshark?

Answer: It indicates TCP has retransmitted data, often because of packet loss or a missing acknowledgement.

96. Can Wireshark capture packets from another computer?

Answer: Normally it captures traffic visible to the selected interface. Other devices' traffic may require suitable network infrastructure and permissions.

5. HTTP Protocol

97. What is HTTP?

Answer: Hypertext Transfer Protocol, an application-layer protocol for communication between web clients and servers.

98. What is HTTPS?

Answer: HTTP secured using TLS encryption and authentication.

99. What is the default HTTP port?

Answer: 80.

100. What is the default HTTPS port?

Answer: 443.

101. What is an HTTP request?

Answer: A message sent by a client to a server requesting a resource or action.

102. What is an HTTP response?

Answer: A message sent by a server in response to a client request.

103. What are HTTP methods?

Answer: They specify the action to be performed on a resource.

104. What does GET do?

Answer: Retrieves a resource.

105. What does POST do?

Answer: Submits data, often to create or process a resource.

106. What does PUT do?

Answer: Replaces a resource or its representation.

107. What does PATCH do?

Answer: Partially updates a resource.

108. What does DELETE do?

Answer: Deletes a resource.

109. What does HEAD do?

Answer: Requests response headers without the response body.

110. What does OPTIONS do?

Answer: Requests information about communication options supported by a resource.

111. What is a URL?

Answer: Uniform Resource Locator; it identifies the location of a resource.

112. What is a requested URL?

Answer: The URL of the resource requested by the client.

113. What is a Request URI?

Answer: The path and optional query portion of the requested resource, such as /online/.

114. What is the Host header?

Answer: It identifies the target host or domain name of an HTTP request, not the client's machine.

115. What are HTTP headers?

Answer: Additional information in an HTTP request or response, such as Host, User-Agent, Accept and Content-Type.

116. What is User-Agent?

Answer: A header that identifies information about the client software, usually a browser or application.

117. What is Content-Type?

Answer: A header that specifies the media type of the message body.

118. What is an HTTP status code?

Answer: A three-digit number indicating the result of an HTTP request.

119. What does status 200 mean?

Answer: OK; the request succeeded.

120. What does status 201 mean?

Answer: Created; a resource was successfully created.

121. What does status 301 mean?

Answer: Moved Permanently.

122. What does status 302 mean?

Answer: Found; a temporary redirect.

123. What does status 400 mean?

Answer: Bad Request.

124. What does status 401 mean?

Answer: Unauthorized; authentication is required or has failed.

125. What does status 403 mean?

Answer: Forbidden; the server refuses the request.

126. What does status 404 mean?

Answer: Not Found.

127. What does status 500 mean?

Answer: Internal Server Error.

128. What does status 503 mean?

Answer: Service Unavailable.

129. What are HTTP status-code classes?

Answer: 1xx informational, 2xx success, 3xx redirection, 4xx client error and 5xx server error.

130. What is a web server?

Answer: Software that handles HTTP requests and sends HTTP responses.

131. What is a web client?

Answer: A browser or application that sends requests to a web server.

132. Is HTTP stateful or stateless?

Answer: HTTP is stateless at the protocol level; each request is independent.

133. What is a cookie?

Answer: A small piece of data stored by the browser and sent to a website when appropriate.

134. Difference between HTTP and HTTPS?

Answer: HTTP does not inherently encrypt communication; HTTPS uses TLS for encryption and authentication.

6. Packet Tracer and Router Configuration

135. What is Cisco Packet Tracer?

Answer: A network simulation tool used to design and test computer networks.

136. What is a router?

Answer: A device that connects different networks and forwards packets between them.

137. What is a switch?

Answer: A device that connects devices in a LAN and forwards Ethernet frames using MAC addresses.

138. Difference between a router and a switch?

Answer: A router mainly connects networks using IP addresses; a Layer 2 switch connects LAN devices and forwards frames using MAC addresses.

139. What is a LAN?

Answer: A Local Area Network covering a limited geographical area.

140. What is a WAN?

Answer: A Wide Area Network connecting networks over large geographical areas.

141. What is an IP address?

Answer: A logical address used to identify a device interface on an IP network.

142. What is IPv4?

Answer: A 32-bit address format represented in dotted-decimal notation, such as 192.168.1.1.

143. What is a subnet mask?

Answer: It identifies the network and host portions of an IPv4 address.

144. What is the default subnet mask of Class C?

Answer: 255.255.255.0 (traditional classful default).

145. What is a default gateway?

Answer: The router interface a device uses to reach destinations outside its local network.

146. What is static routing?

Answer: Manually configuring routes in a router's routing table.

147. What is dynamic routing?

Answer: Routers learn and update routes automatically using routing protocols.

148. Difference between static and dynamic routing?

Answer: Static routes are manually configured and have no routing-protocol overhead; dynamic routes are learned automatically and can adapt to topology changes.

149. What is a routing table?

Answer: A table containing information about available network routes.

150. What does show ip route do?

Answer: Displays the router's IPv4 routing table.

151. What does show ip interface brief do?

Answer: Displays a summary of interfaces, IP addresses and their status.

152. What does show running-config do?

Answer: Displays the active router configuration.

153. What does show ip protocols do?

Answer: Displays information about active IP routing protocols.

154. What does no shutdown do?

Answer: Administratively enables a router interface.

155. What does shutdown do?

Answer: Administratively disables a router interface.

156. What does ping do?

Answer: Tests network reachability using ICMP Echo Request and Echo Reply messages.

157. What is traceroute?

Answer: A tool that identifies intermediate routers along a path to a destination.

158. What is a MAC address?

Answer: A link-layer address used to identify a network interface on a local network.

159. What is ARP?

Answer: Address Resolution Protocol; it resolves an IPv4 address to a MAC address on a local network.

160. What is a default route?

Answer: A route used when no more specific route matches the destination.

161. How do you configure a static route in Cisco IOS?

Answer: Use: ip route destination-network subnet-mask next-hop. Example: ip route 172.16.20.0 255.255.255.0 192.168.1.2

162. What does enable do?

Answer: Enters privileged EXEC mode.

163. What does configure terminal do?

Answer: Enters global configuration mode.

164. What does interface gigabitEthernet 0/0 do?

Answer: Enters configuration mode for that interface.

165. What does copy running-config startup-config do?

Answer: Saves the current configuration as the startup configuration.

7. CRC and Bit Stuffing

166. What is error detection?

Answer: The process of identifying whether data has been corrupted during transmission.

167. What is CRC?

Answer: Cyclic Redundancy Check; an error-detection technique using binary polynomial division.

168. What is the purpose of CRC?

Answer: To detect errors in transmitted data.

169. Does CRC correct errors?

Answer: No. CRC primarily detects errors; it does not itself correct them.

170. What is a generator polynomial?

Answer: A binary polynomial used as the divisor in CRC calculation.

171. What is the CRC remainder?

Answer: The remainder after modulo-2 division; it is appended to the original data.

172. What is modulo-2 division?

Answer: Binary division using XOR operations instead of ordinary subtraction.

173. What is XOR?

Answer: Exclusive OR; the output is 1 when the two input bits differ.

174. What is bit stuffing?

Answer: A technique used to prevent data bits from being mistaken for frame delimiters.

175. How does bit stuffing work?

Answer: In HDLC-style bit stuffing, a 0 is inserted after every sequence of five consecutive 1s in the data.

176. What is bit destuffing?

Answer: The receiver removes the inserted 0 bits.

177. Why is bit stuffing needed?

Answer: It prevents data from accidentally containing the same bit pattern as the frame flag.

178. What is a frame?

Answer: A unit of data transmitted at the data link layer.

179. What is framing?

Answer: The process of dividing a continuous stream of bits into identifiable frames.

8. OSI and TCP/IP Models

180. What is the OSI model?

Answer: The Open Systems Interconnection model is a seven-layer conceptual model explaining network communication.

181. Name the seven OSI layers.

Answer: Application, Presentation, Session, Transport, Network, Data Link and Physical.

182. What is the Application layer?

Answer: It provides network services to user applications.

183. What is the Presentation layer?

Answer: It handles data translation, encryption and compression in the OSI model.

184. What is the Session layer?

Answer: It establishes, manages and terminates communication sessions in the OSI model.

185. What is the Transport layer?

Answer: It provides end-to-end transport, segmentation and related services.

186. What is the Network layer?

Answer: It handles logical addressing and routing.

187. What is the Data Link layer?

Answer: It handles framing, MAC addressing and link-level error detection.

188. What is the Physical layer?

Answer: It transmits raw bits over the physical medium.

189. What is the TCP/IP model?

Answer: A practical networking model used to describe Internet communication.

190. Name the TCP/IP layers.

Answer: Application, Transport, Internet and Network Access (Link).

191. Which OSI layer handles routing?

Answer: The Network layer.

192. Which layer handles end-to-end communication?
Answer: The Transport layer.

193. Which layer handles MAC addresses?
Answer: The Data Link layer.

194. Which layer handles IP addresses?
Answer: The Network layer.

195. Which OSI layer handles encryption and compression?
Answer: The Presentation layer in the OSI model.

196. Which layer handles cables and signals?
Answer: The Physical layer.

197. What is encapsulation?
Answer: Adding protocol headers and sometimes trailers as data moves down the network layers.

198. What is decapsulation?
Answer: Removing headers and trailers at the receiving end.

199. What is a PDU?
Answer: Protocol Data Unit; the unit of data handled at a particular network layer.

200. Name the PDUs at different layers.
Answer: Application: data; Transport: TCP segment or UDP datagram; Network: packet; Data Link: frame; Physical: bits.

9. Important Networking Concepts

201. What is DNS?
Answer: Domain Name System; it resolves domain names to IP addresses.

202. What is DHCP?
Answer: Dynamic Host Configuration Protocol; it automatically provides IP configuration to devices.

203. What is ICMP?
Answer: Internet Control Message Protocol; used for error reporting and network diagnostics.

204. Difference between IPv4 and IPv6?
Answer: IPv4 uses 32-bit addresses; IPv6 uses 128-bit addresses and has a much larger address space.

205. What is localhost?
Answer: A name that refers to the current machine.

206. What is the IPv4 loopback address?
Answer: 127.0.0.1.

207. What is a loopback address?
Answer: An address used by a device to communicate with itself.

208. What is a private IP address?
Answer: An address used inside private networks and not directly routed over the public Internet.

209. Give the private IPv4 ranges.
Answer: 10.0.0.0/8, 172.16.0.0/12 and 192.168.0.0/16.

210. What is a public IP address?
Answer: An IP address that is globally routable on the Internet.

211. What is a subnet?
Answer: A smaller logical division of an IP network.

212. What is bandwidth?
Answer: The data-carrying capacity of a communication link.

213. What is latency?
Answer: The time taken for data to travel from source to destination.

214. What is packet loss?

Answer: The failure of packets to reach their intended destination.

215. What is a firewall?

Answer: A security system that monitors and controls network traffic according to rules.

216. What is a proxy server?

Answer: An intermediary server that forwards requests between clients and other servers.

217. What is NAT?

Answer: Network Address Translation; it translates IP addresses, commonly allowing private devices to share a public IPv4 address.

218. What is a gateway?

Answer: A device or system that connects different networks.

10. Important Port Numbers

219. What are the common FTP ports?

Answer: 20 for data and 21 for control in traditional active FTP.

220. What is the SSH port?

Answer: 22.

221. What is the Telnet port?

Answer: 23.

222. What is the SMTP port?

Answer: 25 (commonly used for server-to-server mail transfer).

223. What is the DNS port?

Answer: 53 (UDP and TCP).

224. What are the DHCP ports?

Answer: UDP 67 for server and UDP 68 for client.

225. What is the HTTP port?

Answer: 80.

226. What is the POP3 port?

Answer: 110.

227. What is the NTP port?

Answer: 123 (UDP).

228. What is the IMAP port?

Answer: 143.

229. What is the HTTPS port?

Answer: 443.

230. What is the SMB port?

Answer: 445.

231. What is the RDP port?

Answer: 3389.

232. What is FTP?

Answer: File Transfer Protocol; used to transfer files between computers.

233. What is SSH?

Answer: Secure Shell; provides secure remote access to another machine.

234. What is Telnet?

Answer: A remote terminal protocol that does not encrypt communication by default.

235. What is SMTP?
Answer: Simple Mail Transfer Protocol; used to send email.

236. What is POP3?
Answer: Post Office Protocol version 3; used to retrieve email.

237. What is IMAP?
Answer: Internet Message Access Protocol; used to access and manage email stored on a mail server.

238. Difference between DNS and DHCP?
Answer: DNS resolves domain names to IP addresses; DHCP automatically provides IP configuration to devices.

11. Linux and Networking Commands

239. What does pwd do?
Answer: Displays the current working directory.

240. What does ls do?
Answer: Lists files and directories.

241. What does cd do?
Answer: Changes the current directory.

242. What does mkdir do?
Answer: Creates a directory.

243. What does rmdir do?
Answer: Removes an empty directory.

244. What does rm do?
Answer: Removes files or directories, with suitable options.

245. What does cp do?
Answer: Copies files or directories.

246. What does mv do?
Answer: Moves or renames files.

247. What does cat do?
Answer: Displays file contents.

248. What does grep do?
Answer: Searches for matching text in files or input.

249. What does chmod do?
Answer: Changes file permissions.

250. What does ps do?
Answer: Displays information about running processes.

251. What does kill do?
Answer: Sends a signal to a process.

252. What does ifconfig do?
Answer: Displays or configures network interfaces; it is a legacy command on many systems.

253. What does ip addr do?
Answer: Displays IP addresses assigned to network interfaces.

254. What does netstat do?
Answer: Displays network connections, routing tables and network statistics.

255. What does ss do?
Answer: Displays socket information and network connections.

256. What does nslookup do?
Answer: Queries DNS servers for domain information.

257. What does ipconfig do?

Answer: Displays network configuration in Windows.

258. What does tracert do?

Answer: Displays the route taken by packets in Windows.

259. Difference between ping and traceroute?

Answer: ping checks reachability; traceroute identifies intermediate network hops.

12. Tricky Viva Questions

260. Can two computers have the same IP address?

Answer: They should not have the same IP address within the same network where addresses must be unique.

261. Can two applications use the same port?

Answer: Generally, they cannot bind to the same IP address and port using the same protocol unless special options or configurations allow it.

262. Can a server communicate with multiple clients?

Answer: Yes. It can use multiple processes, threads or asynchronous I/O.

263. Why is TCP reliable?

Answer: It uses sequence numbers, acknowledgements, retransmissions and error checking.

264. What happens when the server is not running?

Answer: The client connection may fail, commonly with connection refused if the host is reachable but no service is listening.

265. What if the client sends data but the server does not read it?

Answer: Data may remain buffered temporarily; eventually the sender may be blocked or limited by flow control.

266. What does read() returning zero mean on a TCP socket?

Answer: The peer has closed its sending direction (orderly shutdown).

267. What does read() returning -1 mean?

Answer: An error occurred.

268. Why use strlen()?

Answer: It calculates the length of a C string, excluding the terminating null character.

269. What is a null character?

Answer: The character \0 that marks the end of a C string.

270. Why add \0 after receiving data?

Answer: To make received bytes a valid C string before using string functions or %s.

271. What happens if the buffer is too small?

Answer: Data may be truncated; unsafe code can cause a buffer overflow.

272. Why check return values of socket functions?

Answer: To detect errors and prevent incorrect program execution.

273. Difference between a process and a thread?

Answer: A process has its own address space; threads are execution units within a process and share its resources.

274. Difference between a server socket and a client socket?

Answer: A listening server socket accepts connections; a client socket initiates a connection. The server gets a separate connected socket for each accepted client.

275. Why does TCP need sequence numbers?

Answer: To identify byte order and support reliable delivery.

276. What is MTU?

Answer: Maximum Transmission Unit; the maximum packet payload size a link can carry without fragmentation at that layer.

277. What is fragmentation?

Answer: Dividing an IP packet into smaller fragments when necessary to pass through a network with a smaller MTU. IPv4 and IPv6 handle it differently.

278. What is a broadcast address?

Answer: An IPv4 address used to send packets to all devices on a particular subnet.

279. What is unicast?

Answer: One-to-one communication.

280. What is multicast?

Answer: One-to-many communication with a selected group of receivers.

281. What is broadcast?

Answer: One-to-all communication within a broadcast domain.

Quick Reference: Socket Function Sequences

TCP server	TCP client
1. socket() 2. bind() 3. listen() 4. accept() 5. read()/recv() 6. write()/send() 7. close()	1. socket() 2. connect() 3. write()/send() 4. read()/recv() 5. close()

UDP function sequences

UDP server: socket() → bind() → recvfrom() → sendto() → close()
UDP client: socket() → sendto() → recvfrom() → close()

Common Wireshark Display Filters

Purpose	Display filter
HTTP	http
TCP	tcp
UDP	udp
IP address	ip.addr == 192.168.1.10
TCP port 80	tcp.port == 80

Important Cisco Router Commands

Command	Purpose
enable	Enter privileged EXEC mode
configure terminal	Enter global configuration mode
interface gigabitEthernet 0/0	Select an interface
no shutdown	Enable the interface
show ip route	Display IPv4 routing table
show ip interface brief	Show interface summary
show running-config	Display active configuration
copy running-config startup-config	Save configuration
ip route network mask next-hop	Configure a static route

Last-Minute Revision Checklist

- TCP vs UDP differences and three-way handshake
- TCP client and server function sequences
- socket(), bind(), listen(), accept(), connect()
- send(), recv(), sendto(), recvfrom()
- htons(), htonl(), inet_pton(), sockaddr_in
- Wireshark display filters and packet fields
- HTTP methods, Request URI, Host header and status codes
- Static routing and Cisco Packet Tracer commands
- OSI layers and TCP/IP model
- CRC, XOR and bit stuffing
- DNS, DHCP, ICMP, ARP, IP and port numbers
- Linux commands and common socket errors

Question bank contains 281 questions and answers across 12 topics.